

Titanic Survival Analysis Report

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Introduction

This project focuses on analyzing the Titanic dataset to explore factors that influenced passenger survival. Using Python libraries such as Pandas, Seaborn, Matplotlib.pyplot and Scikit-learn, we conducted exploratory data analysis (EDA) and built a logistic regression model to predict survival.

Data Overview

Dataset: Titanic passenger data (<https://www.kaggle.com/c/titanic/models>)

Total rows: 891

Total columns: 12

Target variable: **Survived**

Exploratory Data Analysis (EDA)

Age vs Pclass

We explored the relationship between passenger class and age using boxplots. It was observed that:

1. First-class passengers tend to be older.
2. Third-class passengers were generally younger. This relationship was used to impute missing age values based on passenger class.

Gender Distribution

1. Males had a significantly lower survival rate than females.
2. Gender was converted to a binary column (1 for male, 0 for female) for modeling.

Null Value Analysis

We used a heatmap to visually identify missing values in the dataset.

1. The Age column had around 20% missing values.
2. The Cabin column was missing too much data to be useful(nearly 70%), so we considered dropping it.

Countplot Analysis

Count plots were used to explore the distribution of categorical variables like:

1. Over 550 people didn't survive the Titanic disaster, while a little over 300 did make it.
2. Sex vs Survived (females had a higher survival rate).over 450 men didn't survive the sinking, while only around 100 women didn't make it. On the flip side, more than 200 women survived, compared to about 150 men who made it through.
3. Pclass vs Survived (first-class passengers had better chances of survival).
4. Most passengers were traveling alone, with around 600 having no siblings or spouses aboard. In contrast, about 200 passengers had one sibling or spouse with them

Histplot Analysis

1. The majority of passengers were between the ages of 17 and 35, with noticeably fewer elderly individuals on board.
2. Most people on board paid relatively low fares. Around 400 passengers paid less than 50. As the fare increases, the number of passengers drops quickly. Only a few people paid more than 200 or 300

Pairplot Summary

From the pairplot, it is evident that Pclass and Age show distinguishable patterns in relation to survival.

Data Preprocessing

1. Missing age values were filled based on average age within each Pclass.
2. Categorical columns (Sex, Embarked) were encoded into binary columns.

Model Building

The goal was to predict the **Survived** column — indicating whether a passenger survived (1) or not (0) — using relevant features from the dataset.

Model Used: Logistic Regression

Outcomes

1. The model achieved an accuracy of 82% on the training set.
2. This indicates a strong baseline and confirms that the selected features were meaningful in predicting survival.

Conclusions

EDA Conclusion

1. Class and age had a visible pattern in survival rates.
2. Gender played a crucial role in survival prediction.
3. Data imputation based on class helped maintain accuracy.

Model Conclusion:

Logistic regression achieved a strong accuracy of 82%.The model can successfully predict survival based on simple, interpretable features.Further improvement could be done by scaling the features.